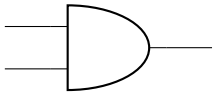
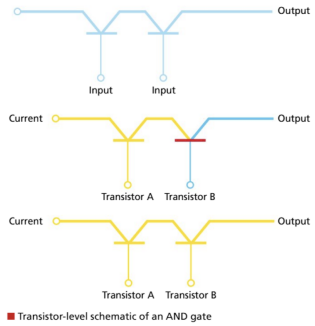
A1.2.3 Purpose and use of logic gates

Logic gates are made up of transistors, which act as electronic switches, allowing or blocking the flow of electrical current. In a transistor, the control wire (or "gate") regulates the current between the two electrodes, known as the "source" and the "drain" . When voltage is applied to the gate, it allows current to flow from the source to the drain, enabling the transistor to switch states and perform logical operations.

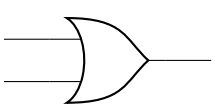


AND
$Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

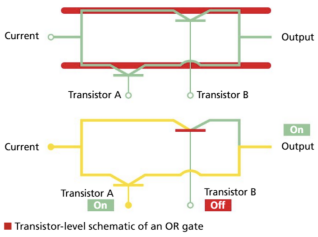


Transistor-level schematic of an AND gate

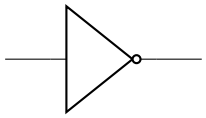


OR
$Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

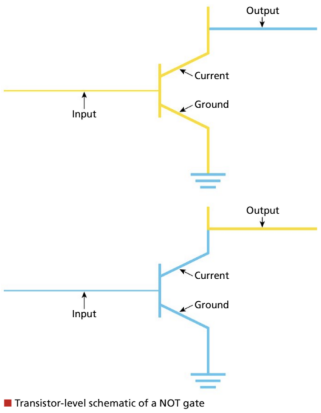


Transistor-level schematic of an OR gate

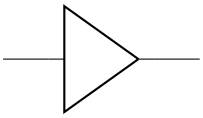


NOT
$Y = \overline{A}$

A	Y
0	1
1	0

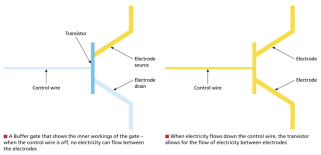


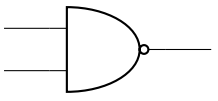
Transistor-level schematic of a NOT gate



BUFFER
$Y = A$

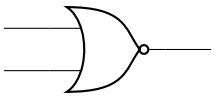
A	Y
0	0
1	1





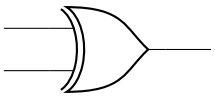
NAND
$Y = \overline{A \cdot B}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



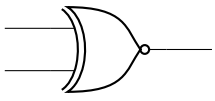
NOR
$Y = \overline{A + B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



XOR
$Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



XNOR
$Y = \overline{A \oplus B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

A1.2.4 Constructing and analysing truth tables

Starting with the intermediate values, work through the logic circuit.
Number of rows in truth table: $2^{\#\text{ of inputs}}$
Worked example: four inputs, all eight gate types.

Truth tables to determine outputs from inputs for a problem description: With a problem description, we first need to identify the inputs. Here we have three: alarm switch, baby crying and room temperature, which we can represent as A, B and C and set up our truth table.

Logical expressions: We can represent logic circuits using Boolean algebra. In Boolean algebra, parentheses are often used to indicate which operations should be performed first. If no parentheses are present, follow this sequence: NOT, AND, OR.

Karnaugh maps and algebraic simplification
Two input K-map: the four cells hold the output for the four possible combinations of A and B. For example, the function $F = A + B$ gives

A \ B	0	1
0	0	1
1	1	1

The two 1s in the column $B = 1$ group to give B, and the two 1s in the row $A = 1$ group to give A, so the K-map simplifies to $F = A + B$.

Three input K-map: the eight cells hold the output for the eight combinations of A, B and C, with the column labels in Gray-code order $00, 01, 11, 10$ so that adjacent cells differ by one variable. For example, the function $F = AB + C$ gives

C \ AB	00	01	11	10
0	0	0	1	0
1	1	1	1	1

The four 1s in the row $C = 1$ (that is, all four columns) group to give C, and the two 1s in the column $AB = 11$ (that is, $C = 0$ and $C = 1$) group to give AB, so the K-map simplifies to $F = AB + C$.

Another three input K-map: the function $F = \overline{A}BC + A\overline{B}C + ABC$ gives

C \ AB	00	01	11	10
0	0	0	0	0
1	0	1	1	1

The two 1s in the row $C = 1$ with $B = 1$ (that is, 01 and 11) group to give BC, and the two 1s in the row $C = 1$ with $A = 1$ (that is, 11 and 10) group to give AC, so the K-map simplifies to $F = BC + AC$.

Grouping conventions

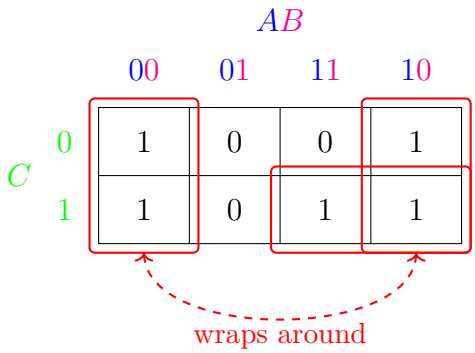
- Groups must contain powers of 2: One, two, four, eight or sixteen 1s can be grouped together.

- Groups must be rectangular or square: Each group should form a rectangle or square shape.
- Groups cannot be diagonal: Adjacent 1s can only be grouped horizontally or vertically, not diagonally.
- Groups must be as large as possible: Always aim to make the largest groups to simplify the expression further.
- Groups can overlap: Some 1s may be included in more than one group if it helps form larger groups.
- Minimize the total number of groups: The goal is to use the smallest number of groups to cover all the 1s.

To determine the expression from the groups, we look at each group of cells and refer to the variables. If a variable' s value stays the same across all the cells in the group, we keep that variable in the simplified expression. Iowever, if the variable' s value changes across the group, we discard it from the expression.

Wrapping around edges in K-maps

K-maps are considered three-dimensional, and groups can be formed from left to right and top to bottom (although only left to right is possible with three inputs). In the example of $F = \overline{B} + ABC$:



The four 1s in the columns $AB = 00$ and $AB = 10$ wrap around the left and right edges to form one group of four in which $B = 0$ throughout, giving $\overline{B}$; the remaining 1 groups with the adjacent 1 in $AB = 10$ (overlapping the $\overline{B}$ group) to give AC, so the K-map simplifies to $F = \overline{B} + AC$.

Four-variable K-maps wrap in both directions, so the map behaves like a torus: the left and right edges join, and the top and bottom edges join as well. The figure shows a 4×4 map and the same map drawn on a torus; the four wrapping corner cells are marked with dots.

